

Design Thinking Stages Final Documentation

Group 04 – LogiTeh

furlink

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Stage 1: Empathize

The first phase of the design thinking process is to have a deep understanding of the target users and to identify the problems they have at hand. The team lay aside all their assumptions about the problem, and the target users. This allows them to objectively consider all possibilities for the customers and their needs.

1.1 Client and Customer Interviews

furlink is a startup project being built by a student team. The team did not work with a specific client, instead, the team conducted market research to understand the needs and challenges within the pet industry in Makati City, Philippines. A quantitative descriptive research design was implemented to understand the booking preferences and needs of pet owners and service providers for a centralized pet booking platform. Data was collected using a Likert scale survey questionnaires distributed online (Microsoft Forms via targeted social media ads) and onsite. The team also held online interviews for pet grooming shops located in Makati City. This approach allowed the team to identify pain points, gaps, and opportunities for innovation.

Among the sixty-four (64) respondents from the survey conducted for pet owners, forty-three (43) were identified as pet owners from Makati City, and interviews were held online with owners of pet grooming shops located in Makati City.

The following are the questions the group has made for the pet owners:

Question	Answer
As a fur parent, what problems do you encounter when	Long waiting time, fully booked grooming shops, uneven haircut, unmet haircut expectation, no advance booking, and overlapping schedules

availing pet grooming services?	
Do you still encounter problems even though you have already booked a grooming service for a specific date and time?	Schedule conflicts, double booking, sudden cancellation of and appointment

The answers are generalized and derived from the survey.



Figure 1 Interview with The Groom Room by Steve

The team was able to interview the owner of The Groom Room by Steve where they were able to identify pain points from a pet grooming shop owner's perspective.

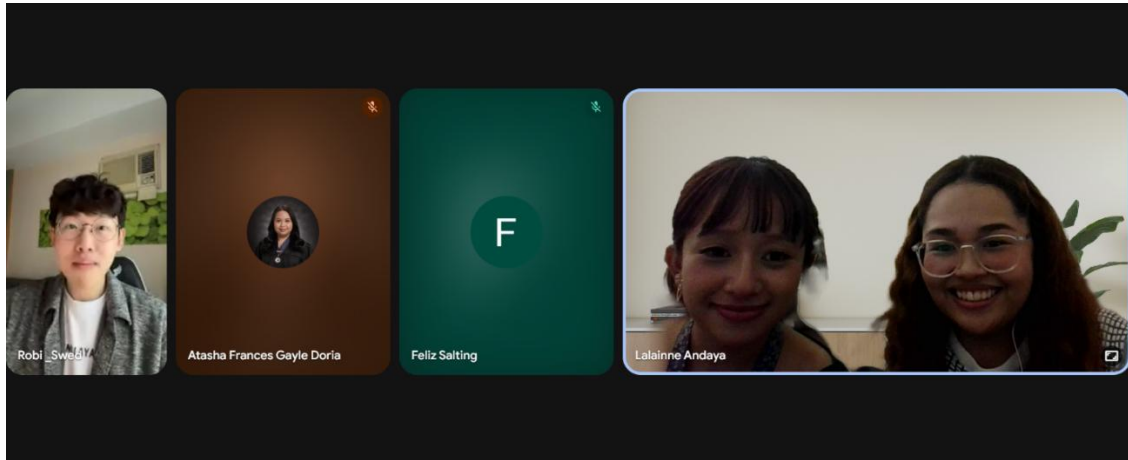


Figure 2 Interview with The Nature Pet Grooming Salon

The team was also able to interview the owner of the pet grooming shop, The Nature Pet Grooming Salon. The owner had expressed their need for a way to market their shop since it only recently opened. Furthermore, the owner had also expressed their need for a conditional waiver to protect both their staff and the pet owners.

The following are the questions the group has made for the pet owners:

Question	Answer
Do you allow your customers to book in advance?	Answers from the grooming shop varied between allowing advance booking and not allowing bookings.
How do you handle or organize your customers' bookings?	Through a drawn calendar on a whiteboard, Excel, and notebook.
How do you handle walk-in customers when you can't accommodate them?	Politely ask them to wait or go back on a later time.

How do you think a booking system will help you handle or organize your bookings?	It will help manage bookings, and it will let us see the grooming schedules better.
What challenges do you encounter in your business in terms of accommodating your customers?	Too many customers to handle during peak season like summer and pet groomers sometimes get bitten by pets and pet owners do not take responsibility. Some also answered that there was a lack in customer because they were a newly opened shop.

The answers are generalized and derived from the survey.

1.2 5 WHYS (Simplified)

Factor 1: Long waiting time or delayed service

1. Why are pet owners waiting too long?
 - Grooming shops are overbooked.
2. Why are grooming shops overbooked?
 - Bookings are handled manually, leading to scheduling conflicts.
3. Why is the booking process manual?
 - The shops do not use scheduling tools.
4. Why do grooming shops not use scheduling tools?
 - Grooming shop owners believe it is too costly and complicated to implement.
5. Why do they believe it's costly and complicated to use?
 - They are unaware of available affordable and easy-to-use booking platforms.

Root Problem: The manual process of tracking bookings and the rising customers during peak seasons. Because of the rising number of customers, grooming shops tend to struggle keeping up with schedules. They are unaware of available online booking systems that are efficient and free.

Factor 2: Miscommunication about service

1. Why is the pet owner dissatisfied?
 - Haircut styles are different from what they asked for.
2. Why were the haircut styles different?
 - The pet groomers often misunderstand instructions.
3. Why do their pets receive different haircut styles?
 - Instructions tend to be given verbally without any reference image.
4. Why were there no reference images?
 - Some pet owners do not know what haircut styles are called, and pet shops do not specifically ask for the images,
5. Why do grooming shops not ask for reference images?
 - They believe reference images are not a requirement.

Root Problem: There is no standardized and documented process when it comes to confirming booking requests and grooming preferences. Pet groomers rely on verbal instructions and most of the time, they do not meet eye-to-eye with pet owner's requests, or if they do, they forget due to the number of pets they accommodate.

Factor 3: Lack of Waivers and Terms and Conditions

1. Why are disputes occurring?
 - There is no written agreement on liabilities.
2. Why is there no agreement?
 - The shops have no waiver or terms and conditions document.
3. Why do pet grooming shops not have one?
 - They do not know how to create one or they are unaware that waivers are needed.
4. Why do they do not create one?
 - Pet grooming shops do not see it as a priority.
5. Why are waivers not seen as a priority by pet grooming shops?
 - They underestimate the benefits of having waivers.

Root Problem: The lack of formal and written agreement that outlines the scope of the service, risks, and liabilities between pet owners and pet grooming shops. This results in both parties being vulnerable in case of disputes, injuries, or accidents.

1.3 Actual Personas

Pet Owner Profile

Name: Carleen

Age: 20 years old

Gender: Female

Pet/s: Cats and dogs

Service/s Available: Full Grooming, bath and blow dry, and other pet grooming services

Pet Grooming Service Provider Profile

Business Name: The Groom Room by Steve

Location: Makati Avenue, Makati City

1.4 Theoretical Personas

PERSONA 1: ANICA

Office worker

Dog owner of a 20kg Aspin breed

Does not know where to find reliable and quality grooming services

PERSONA 2: KENDALL

Retired lawyer who just opened a pet grooming shop

Is not tech-savvy

Manually organizes bookings

PERSONA 3: LOGAN

Manager at a company

Dog owner of a 7-year-old golden retriever

Finds it difficult to communicate with the pet groomer about his desired haircut for his pet

PERSONA 4: RAMON

Grad student

Cat owner of a 3-year-old Persian cat

Wants an efficient online pet grooming service booking system

PERSONA 5: BEA

Pet grooming shop owner

Located in an area with a lot of competitors

Finds it difficult to sustain loyal customers

1.5 Customer Interviews (Simplified)

PET OWNERS

Carleen Zarsuela Samillano

As a fur parent, what problems do you encounter when availing pet grooming services?

- Scheduling conflicts
- Long queues
- Incorrect haircuts do not match their requests.
- Groomer unavailability, and pets experiencing trauma from mishandling
- Fully booked services

Do you still encounter problems even though you have already booked a grooming service for a specific date and time?

- Yes, I still experience problems including booking overlaps, having to wait despite confirmed appointments, scheduled times not being honored.

PET GROOMING SERVICE PROVIDER

The Groom Room by Steeve

Location: Makati City

Do you allow your customers to book in advance?

- Yes, we allow customers to book in advance.

How do you handle or organize your customers' bookings?

- We accept bookings through Facebook and Instagram, and we track them using Google Calendar.

How do you handle walk-in customers when you can't accommodate them?

- We give them option to book on a different day that works for them.

How do you think a booking system will help you handle or organize your bookings?

- It would help speed up the appointment processing, all details required would be answered, and it will establish the business in the market.

1.6 Empathy Map

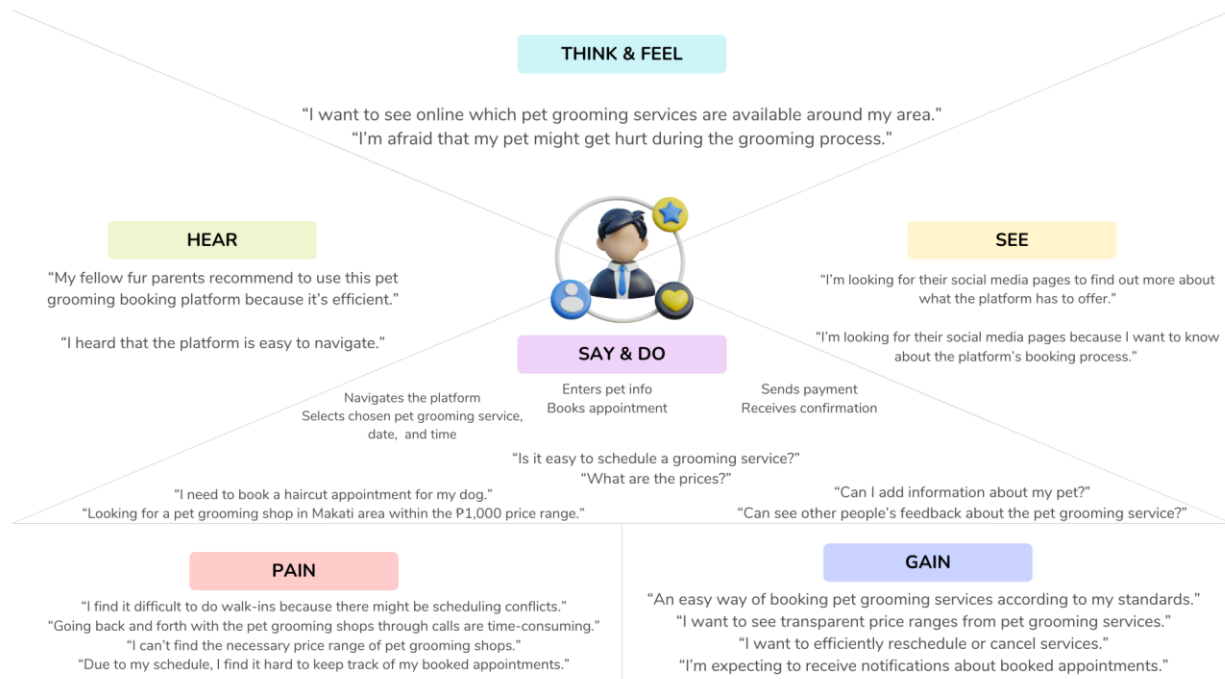


Figure 3 Empathy Map

1.7 Pain-Gain Analysis

Pain

1. What are the customers' frustrations?
 - Scheduling conflicts with the pet service providers
 - Hassle of finding the necessary price ranges of pet services
2. What prevents the customers from being satisfied?
 - Going back and forth with the service providers about scheduling and information is time consuming.

Gain

1. What are the customers' expectations?
 - Hassle free booking

- Quickly reschedule or cancel a booking
- Transparent view of each pet service provider's price range

2. What satisfies the customers?

- "An easy way of booking of booking pet grooming services according to my standards."
- "I want to see transparent price ranges from pet grooming services."
- "I want to efficiently reschedule or cancel services."
- "I'm expecting to receive notifications about booked appointments."

Analysis

- To have an online booking platform for pet service providers.
- Provide transparency of price ranges that pet service providers offer.
- To have a way of browsing pet service providers easily.
- To have a feature that updates the in-app calendar in real-time to avoid scheduling conflicts.

Stage 2: Define

The second phase of the design thinking process is by putting together all the information gathered from the previous phase. This stage is to define the problem statements clearly which should be based on human-centered needs, rather than being focused on the business goal.

2.1 Clustered Problems

The group determined the problems and grouped them into three parts: business process, online selling and marketing, and general problems. This is also based on the root problems the group has found out during stage 1. The team utilized Microsoft Whiteboard to visualize the clustering of problems.

Booking Process:

- Missed Appointments
- Double booking
- No waiver for liabilities
- Lack of business report
- Long waiting time

Service Information:

- Lack of customer
- Unstandardized market pricing
- Customer's lack of trust
- Lack of feedback from past customers

Service Management:

- Struggle to capture customers' desired haircut style for their pets.
- No automated calendar view that tracks confirmed booking.
- Having to manually contact customers through SMS or social media accounts.
- After grooming pets might have grooming wounds or scratches.
- Pet service providers sending wrong notifications to pet owners.
- Unorganized schedule



Figure 4 List of Problems Whiteboard Screenshot



Figure 5 Sorted Problems Whiteboard Screenshot

2.2 How-Might-We Reimagine



Figure 6 How Might We Whiteboard Screenshot

2.3 Stage 2 Design Process

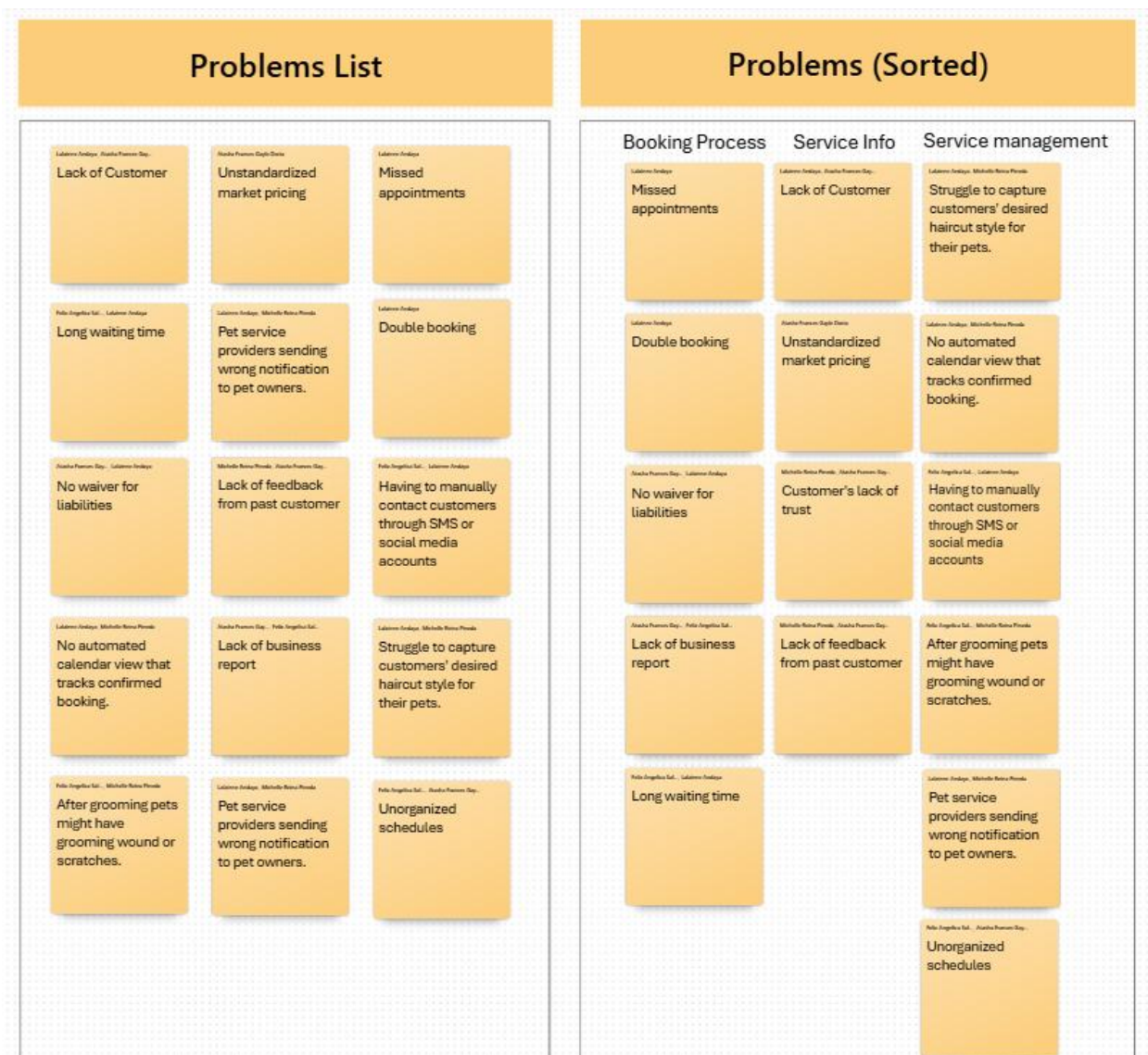


Figure 7 Design Process Whiteboard Screenshot

Stage 3: Ideate

On the third phase of the design thinking process, the team started brainstorming ways to meet the needs of their target users. The team gathers with open minds to come up with as many possibilities as they can and investigate and test them by the end.

3.1 Brainstorming Session

In this session, the group discussed the potential features and solutions for the web application.

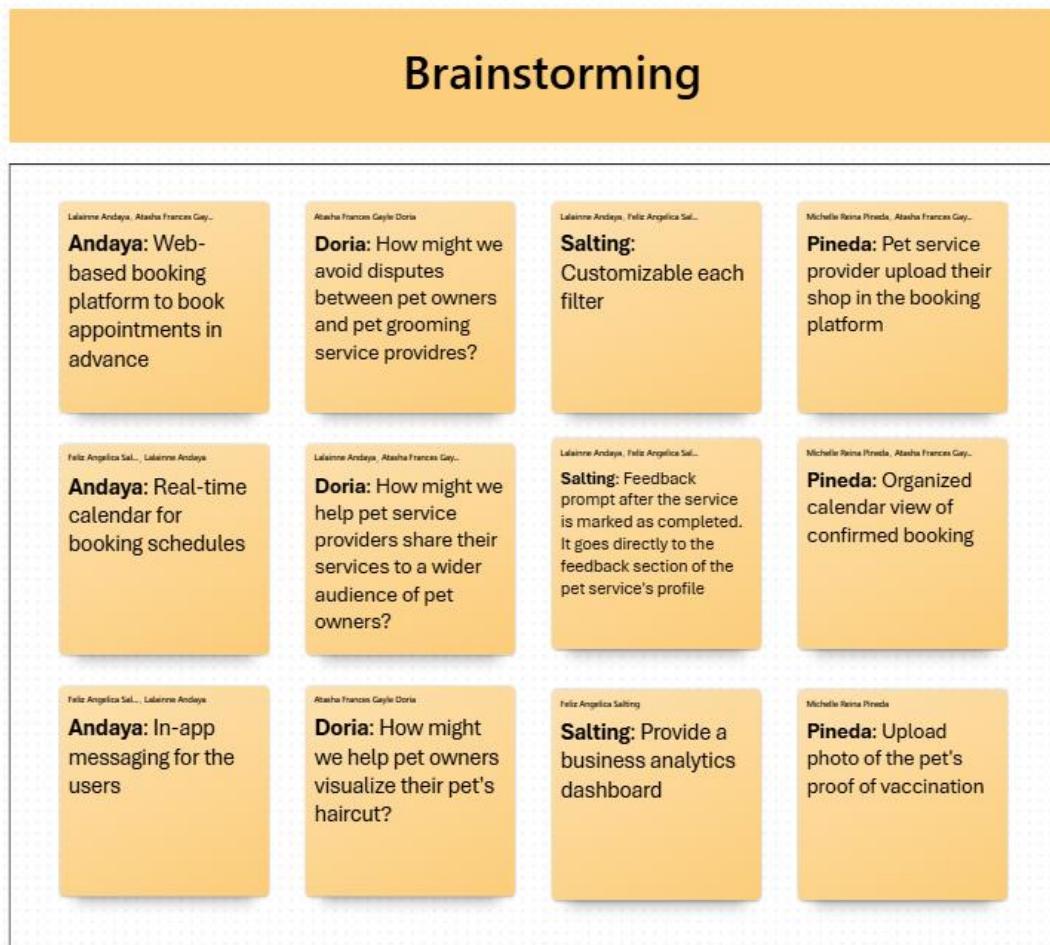


Figure 8 Brainstorming Whiteboard Screenshot

3.2 Crazy 8's

Crazy 8's is a fundamental Design Sprint method. It is a quick sketching exercise that requires participants to sketch eight distinct ideas in eight minutes.

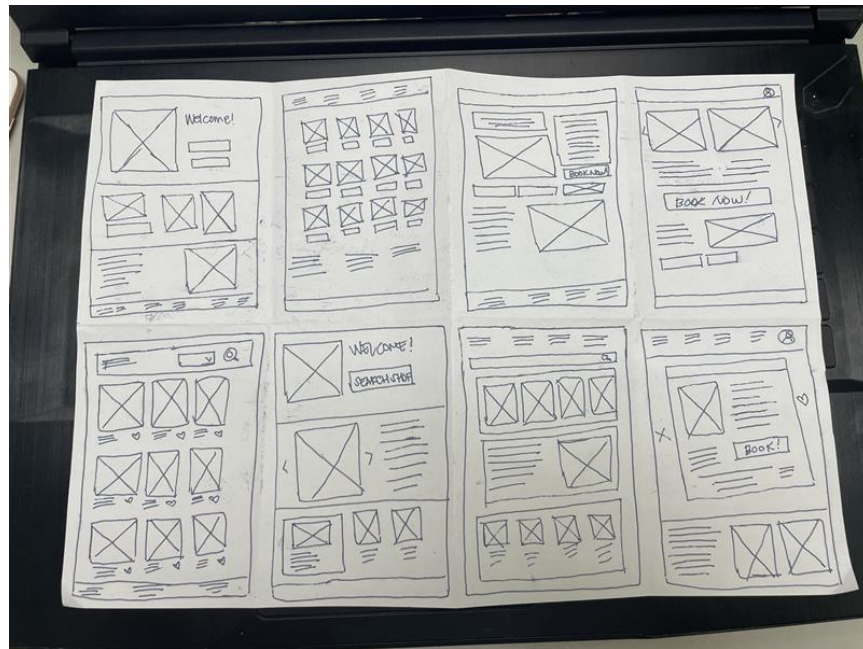


Figure 9 Crazy 8s Andaya

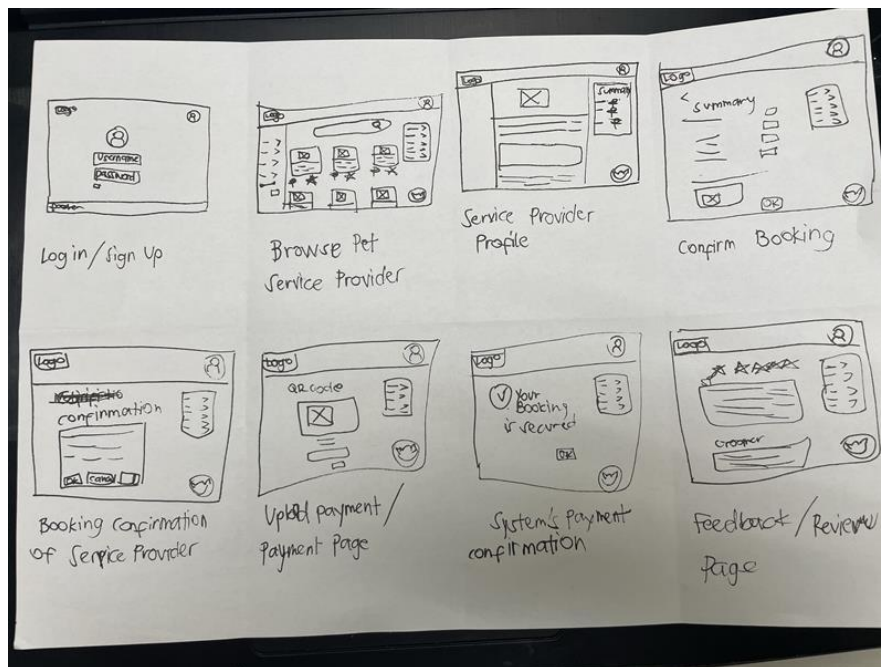


Figure 10 Crazy 8s Doria

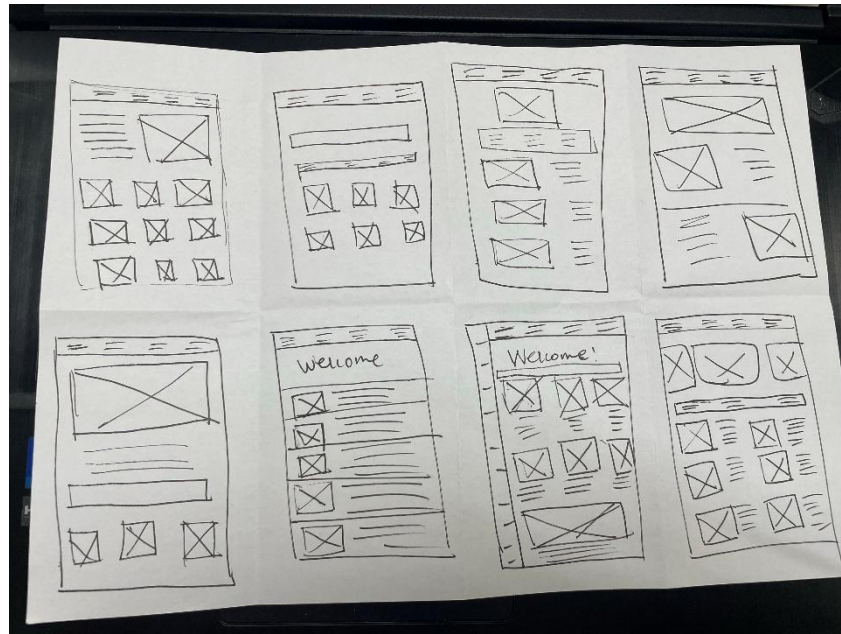


Figure 11 Crazy 8s Pineda

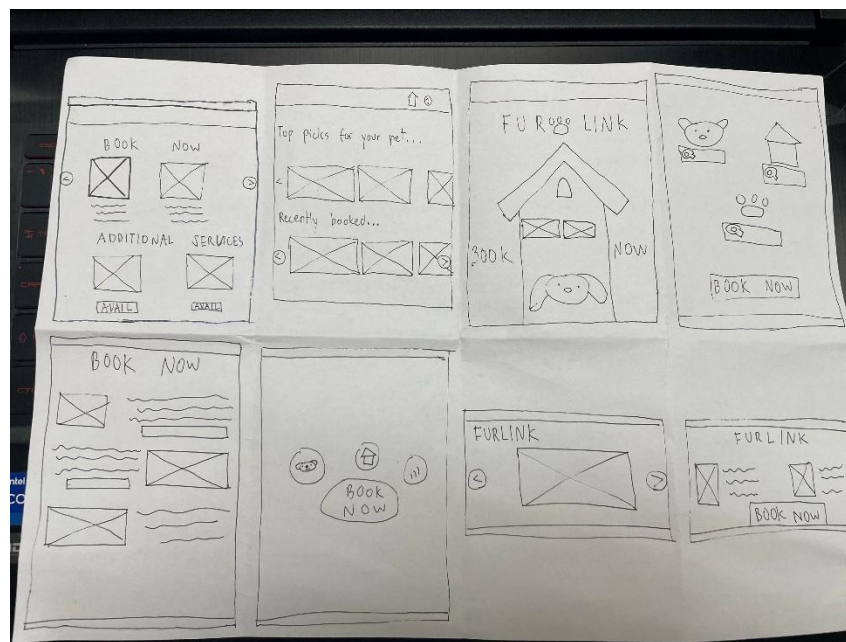


Figure 12 Crazy 8s Salting

3.3 Post-it Voting

Dot Voting is a simple decision-making and prioritization technique in UX design: Participants in UX workshops, activities, or collaborative sessions vote on the importance of design ideas, features, usability findings, and anything else that requires prioritization by placing colored dots.

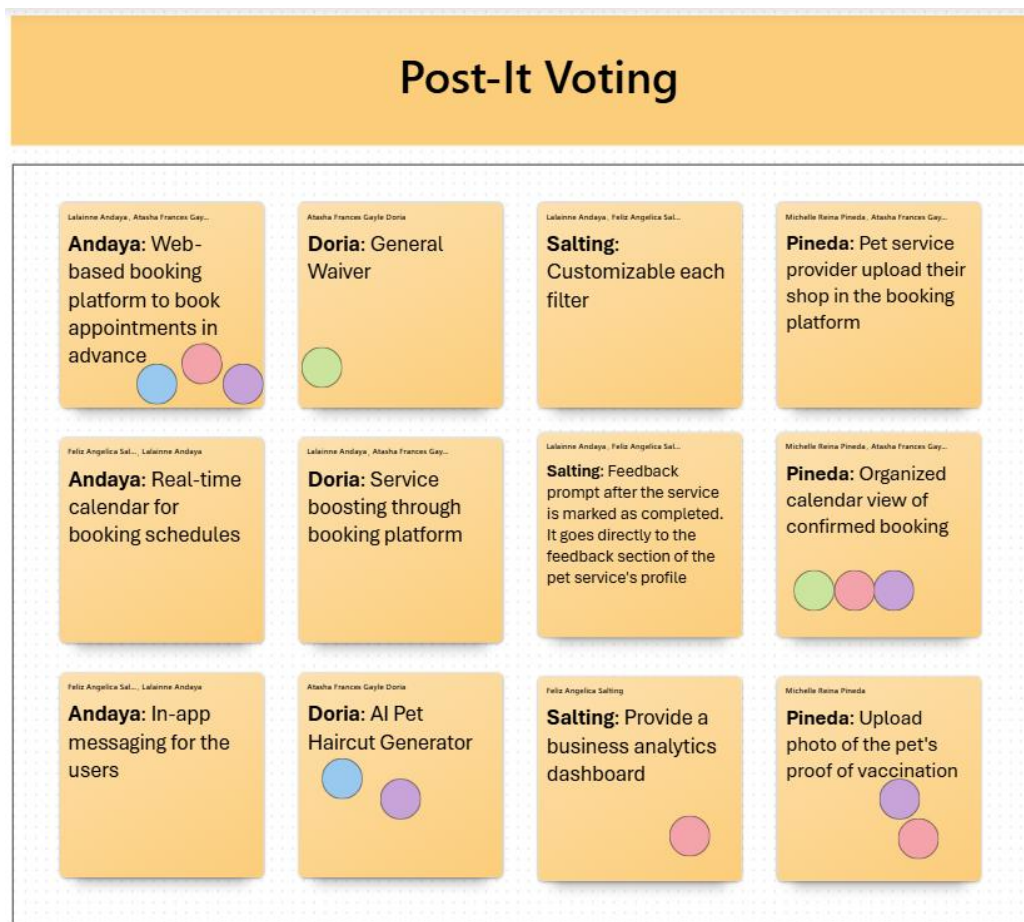


Figure 13 Post-It Voting Whiteboard Screenshot

3.4 Desirable, Viable, Feasible

The ideal innovation process combines desirability, feasibility, and viability, as follows: Are we addressing the correct pain point? A desirability test determines whether your solution is a nice to have or a must have for your customer. Are we capitalizing on our core operational strengths?

A feasibility test evaluates the operational capabilities of a new solution. Is our solution viable? Does it contribute to long-term growth? The final test for your innovation focuses on your solution's value chain to ensure that it is viable now and in the future.



Figure 14 Desirable, Viable, Feasible Whiteboard Screenshot

Stage 4: Prototype

One of the most important stages in the design thinking process is prototyping. Through prototyping, the ideas from the previous stage is turned into tangible forms, allowing exploration and feedback.

4.1 Storyboard

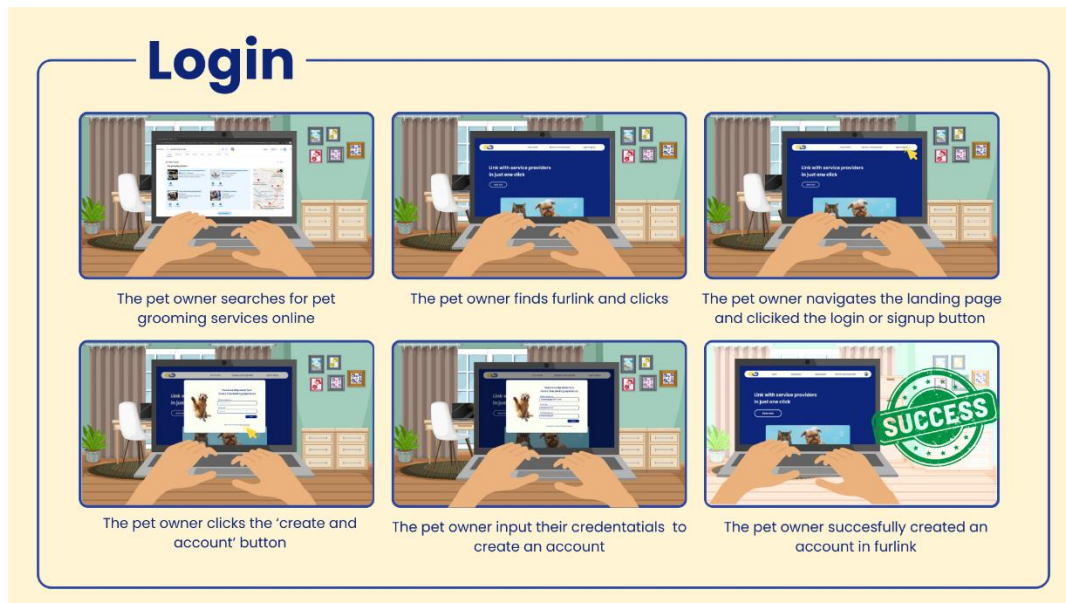


Figure 15 Storyboard: Login

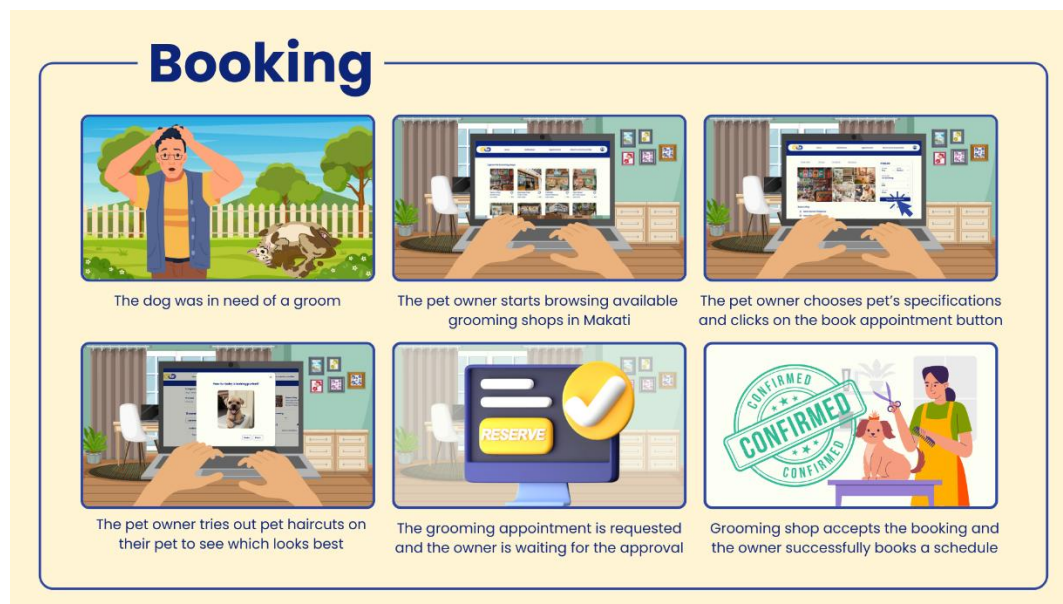


Figure 16 Storyboard: Booking

Posting



The business owner is overwhelmed by a flood of customer booking requests.



His staff recommended a staff member recommends using furlink.



His staff explains that furlink allows providers to post their services and accept bookings.



The owner starts posting his services on the platform.



New bookings start rolling in and are now perfectly organized.



After some time using furlink, his thriving business has organized bookings.

Figure 17 Storyboard: Posting

Payment Confirmation



The grooming shop receives a new booking request



The grooming shop accepts the request



The pet owner receives a notification that the booking is accepted



The pet owner chooses a payment option to pay the downpayment



The owner scans the QR, sends the payment, and answers the form



The grooming shop confirms that payment has been accepted

Figure 18 Storyboard: Payment Confirmation

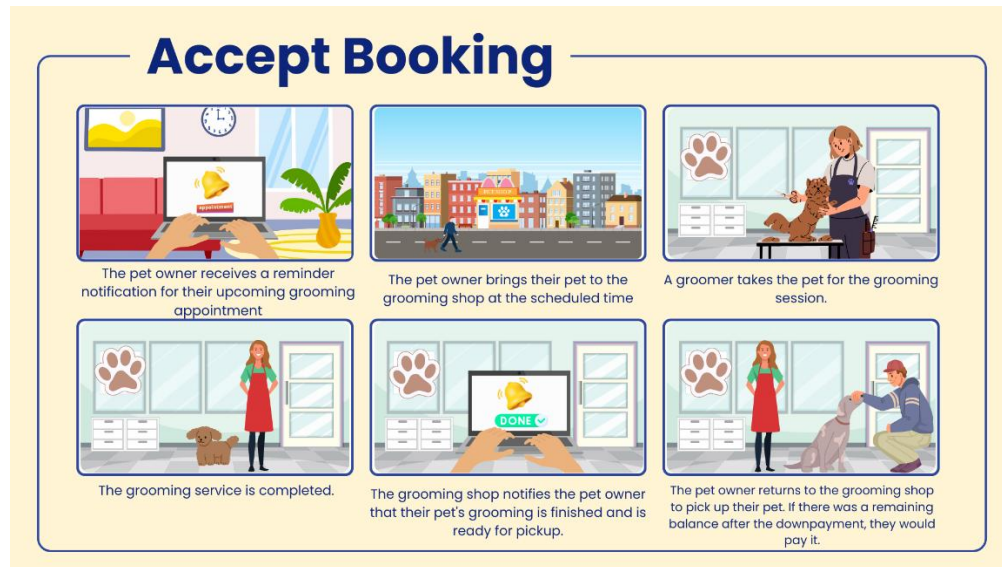


Figure 19 Storyboard: Accept Booking

4.2 Wireframe

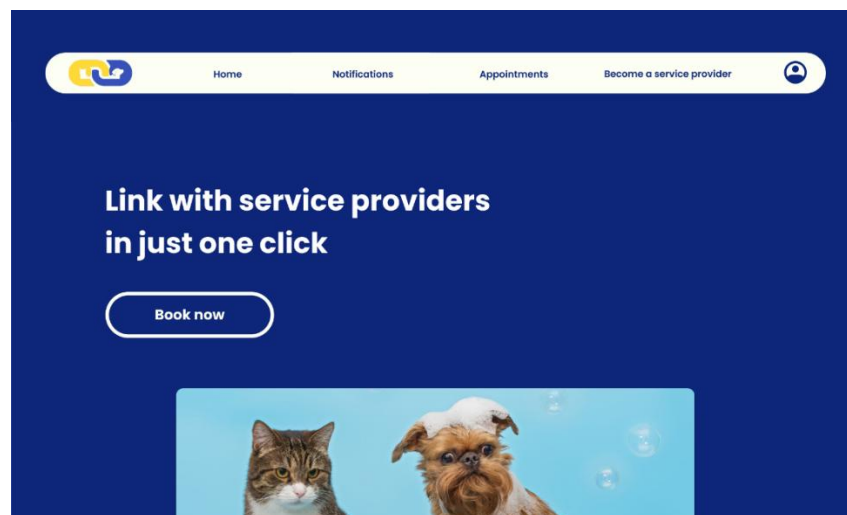


Figure 20 Wirefram Homepage

Figure 20 is *furlink*'s home page. The users will be welcomed with this page when they are not logged in to their account. Through the home page's header, users can explore what *furlink* is or log in or signup. Other features such as *Become a service provider*, and the booking feature can only be accessed by users if they are logged in.

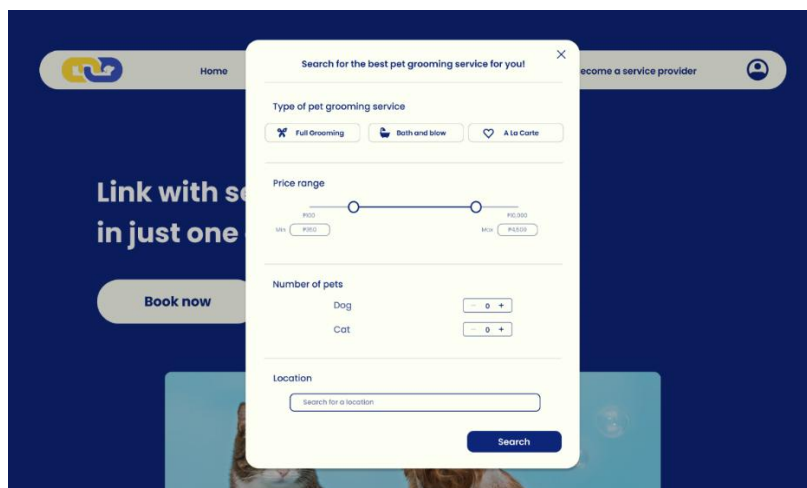


Figure 21 Filter Dog or Cat Grooming Services

The above figure shows a modal that shows up when the users have successfully logged in, and when they click the *Book now* button. The modal lets users filter dog or cat grooming services based on their preferences, and upon clicking the *Search* button, the users are brought to a new page where they are shown filtered grooming services.

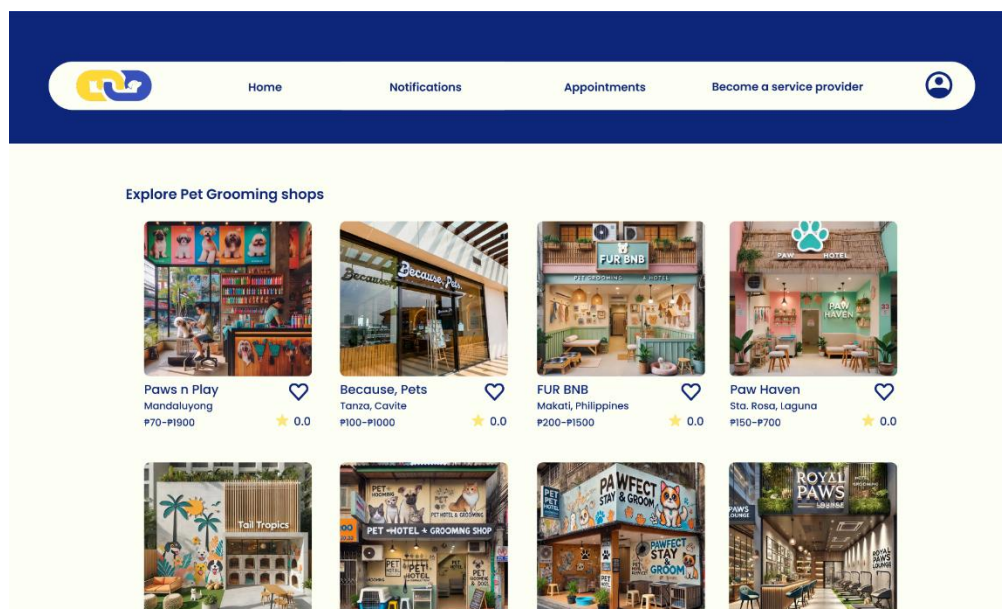


Figure 22 Explore Dog or Cat Grooming Shops Page

Figure 22 is the page shown after the pet owner filters grooming services through the filter modal. In this page, they can explore dog or cat grooming shops that were filtered based on their preferences and choose the right shop for their pets.

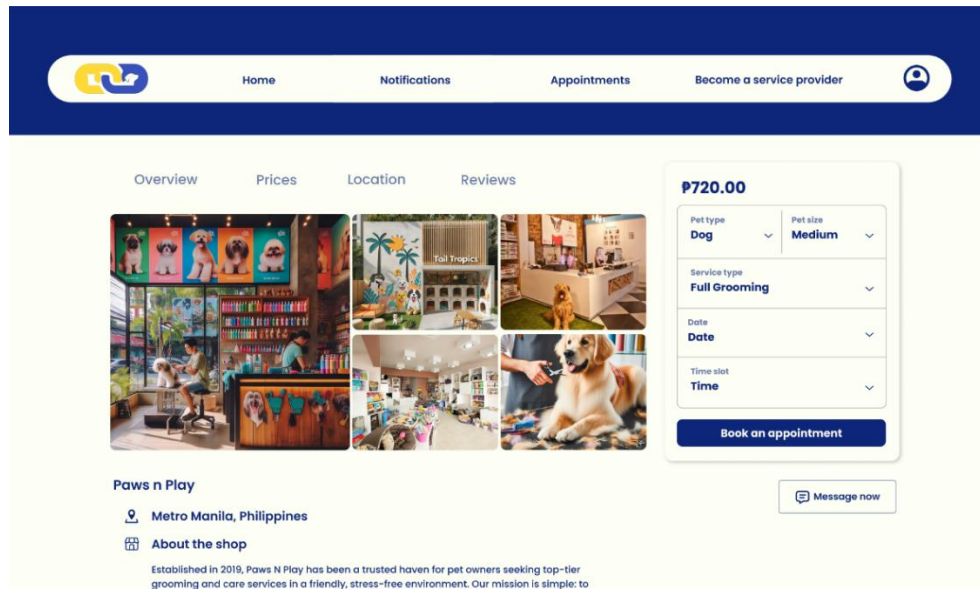


Figure 23 Book an Appointment Page

The above figure shows a dog or cat grooming shop the pet owner clicked on. It has necessary information about the shop such as an overview about the shop, their prices, the location, and reviews left by previous bookers. The pet owner can also reach out and message the shop through the webapp's in-messaging feature. Furthermore, they can adjust their appointment based on their pet's information and their preferred date and time.

The screenshot shows a web application interface for booking a grooming appointment. The top navigation bar includes links for Home, Notifications, Appointments, and Become a service provider. The main content area is titled 'Book an appointment' and displays the following details:

- Date:** April 22, 2025 (with an 'Edit' link)
- Service:** Full Grooming Service (with an 'Edit' link)
- Pet type and size:** Dog, 1 Medium (with an 'Edit' link)
- Time slot:** 9:00 AM (with an 'Edit' link)

To the right, a summary box for 'Paws n Play' (Barangay San Rafael, Mandaluyong City, Metro Manila, Philippines) shows:

- Full Grooming Dog Medium x1** at **₱720.00**
- Total** at **₱720.00**
- A link to 'See price breakdown'

Below the booking details, there is a section for 'Grooming Specifications' with a button to 'Add reference photo' and a text area for 'Additional notes to Grooming shop'.

Figure 24 Booking Information Summary Page

Figure 24 is a page that shows the dog or cat owner's booking summary information. It contains the date and time of the appointment, the pet information, and the total price and price breakdown of their availed service. In this page, the pet owner can also upload and have an image of their preferred haircut style on their pets, and add additional notes to the grooming shop. Once the pet owner has confirmed every information, and is satisfied, they can click on the Confirm Booking Appointment button to send the booking request.

The screenshot shows a web application interface for selecting services. The top navigation bar includes links for About furlink and Switch to customer. The main content area is titled 'Which service' and includes a button for 'Full Grooming'. A modal window titled 'Full Grooming' is open, displaying pricing information for dogs and cats.

For Dogs

Size	Kilo	Rate
Small	> 7kg	₱500.00
Medium	7kg - 10kg	₱600.00
Large	10kg - 20kg	₱700.00
Extra Large	20kg - 30kg	₱800.00

For Cats

Size	Kilo	Rate
Small	> 4kg	₱500.00

The modal also includes an 'Add more' button and a 'Next' button.

Figure 25 Service Listing

Figure 25 displays the service listing page for a logged-in pet grooming service provider. Service providers initiate the process by entering price lists for each service. After listing services, they can submit important business details including name, service overview, location, facility photos, payment QR code, waiver, business permit, and certifications.

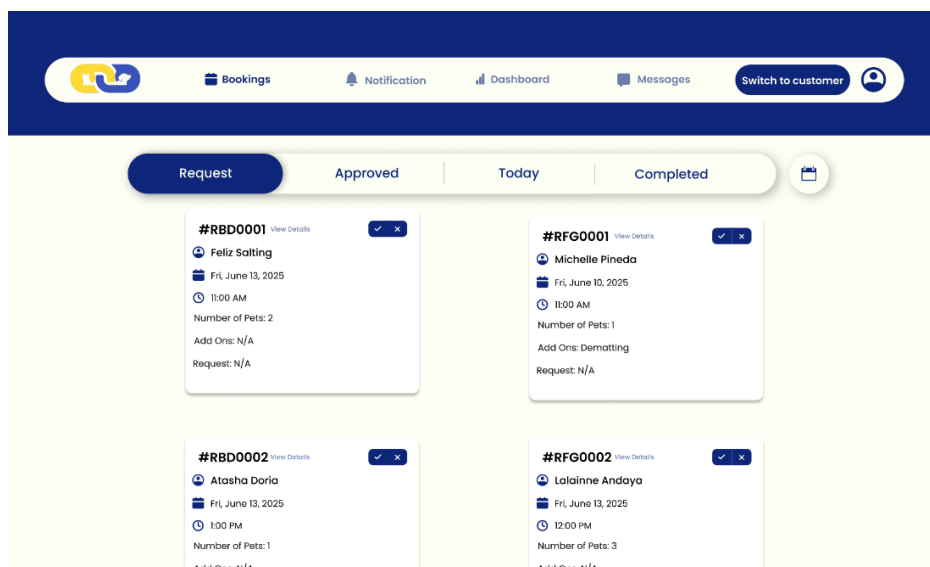


Figure 26 Booking Page

The booking page displays all incoming booking requests received by service providers, offering options to either accept or decline each one. This same page also allows providers to view their approved bookings, today's confirmed bookings, and successfully completed bookings.

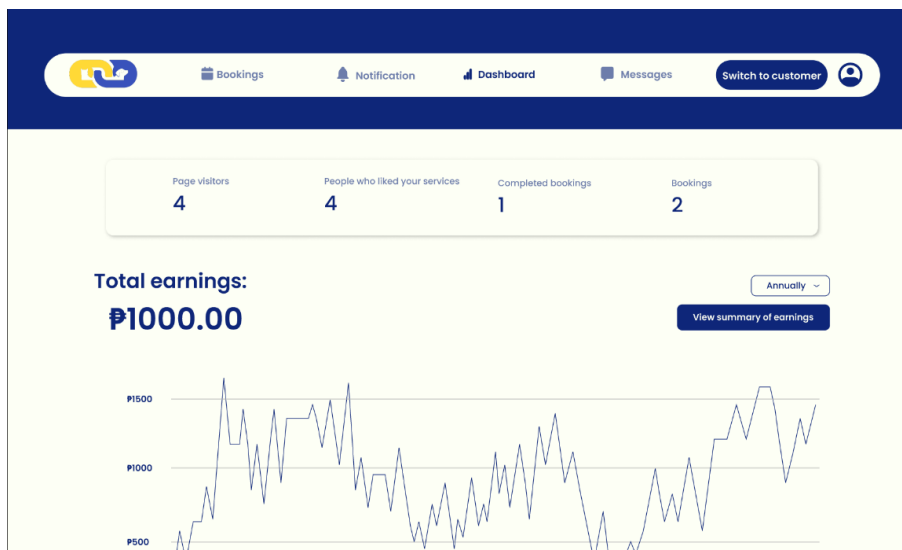


Figure 27 Dashboard

The dashboard page provides a summarized overview of key metrics, including page visitors, service likes, completed bookings, and total accepted bookings (which includes completed bookings) within the platform. Additionally, it features a line graph enabling service providers to visualize their earning trends alongside their total earnings.

Stage 5: Test

The last phase of the design thinking process involves testing the solutions identified in the Prototype phase. This is the final stage of the process, but the results on this stage are used to redefine one or more problems. In this stage, pet owners evaluated the prototype by booking a grooming service and exploring the webpage. Furthermore, the grooming shops interviewed in the first phase of the process were also able to evaluate and simulate posting the services they offer.

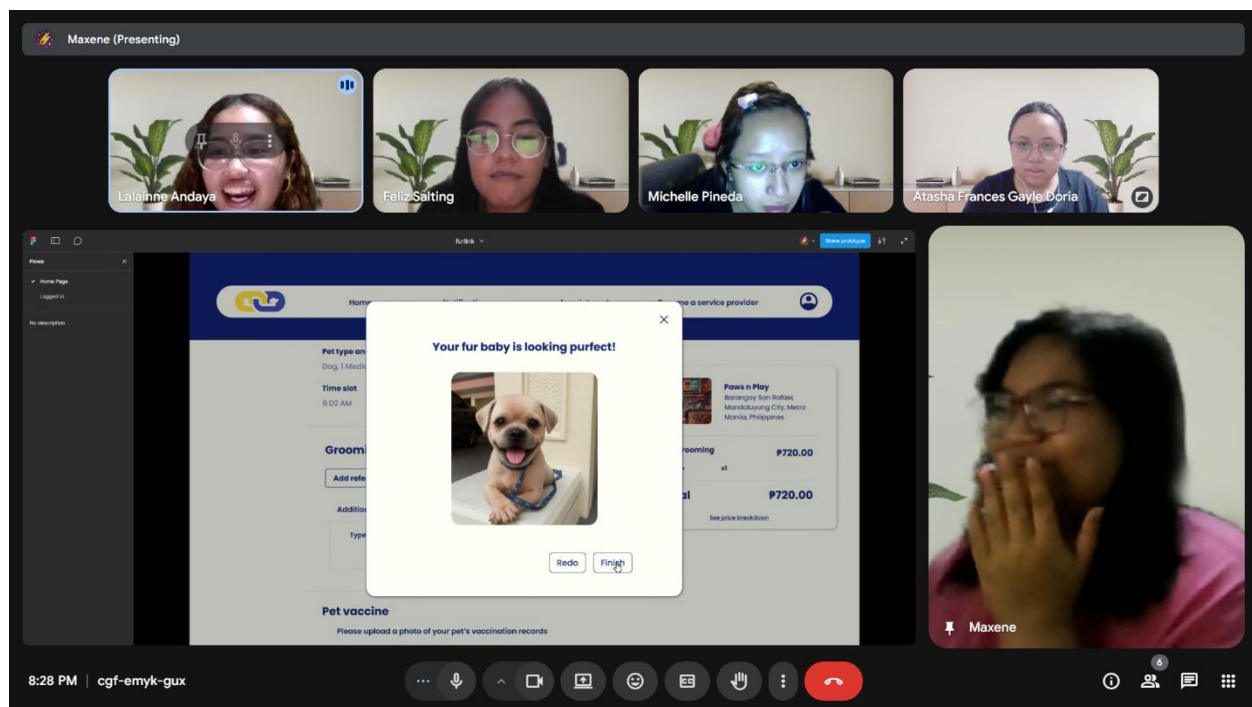


Figure 28 Feedback 1.0



Figure 29 Feedback 2.0

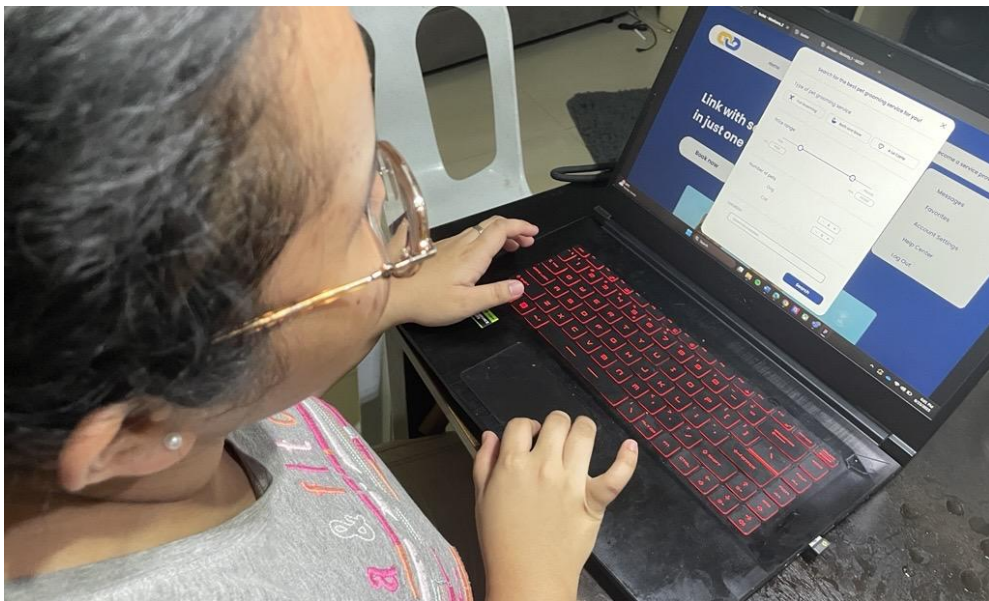


Figure 30 Feedback 3.0

Feedback Summary

Overall, the testers liked the UI/UX of the prototype and the color scheme was the perfect shade of blue, it was neither too bright nor too dull, making it easier for the eyes. The font and font size used were the perfect size and the texts were easily readable. The testers also expressed their liking on the dashboard feature but commented on adding other graphs instead of just a line graph. They also liked the calendar feature that shows the service providers their schedules and bookings.

On the other hand, there was feedback for the system's improvement such as, too static, a hamburger menu could be used instead of the current tab usage, and to use the yellow to maximize the color scheme. Testers also mentioned that the system should highlight the liabilities and make sure that the users are well-aware that furlink is just a third-party booking system and is not liable for any risks and disputes. Additionally, the system should highlight that furlink does not process any payments. All payments happen between the pet owner and the service provider, and downpayments are non-refundable. They did mention that it would be better if we had a payment gateway for future releases to make the webapp more reliable. Another feature they recommended was to let pet owners view a shop's pet groomer profiles for reliability and add what the system does to the image they upload in the AI haircut generator.